

Submitted for Publication to *J. Bio Inorg Chem*
August 27, 2017

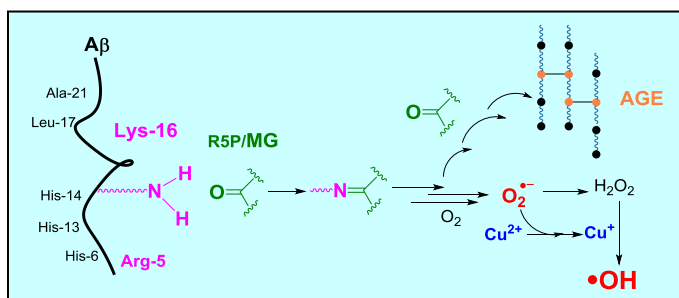
Glycation of Lys-16 and Arg-5 in Amyloid- β and the Presence of Cu^{2+} Play a Major Role in the Oxidative Stress Mechanism of Alzheimer's Disease

Sebastian M. Fica-Contreras, Sydney O. Shuster, Nathaniel D. Durfee, Gregory J. K. Bowe, Nathaniel J. Henning, Staci A. Hill, Geoffrey D. Vrla, David R. Stillman, Kelly M. Suralik, Roger K. Sandwick, and Sunhee Choi*

Department of Chemistry and Biochemistry, Middlebury College, Middlebury, VT 05753

*Author to whom correspondence should be addressed: E-mail: choi@middlebury.edu.

ABSTRACT



Extensive research has linked the amyloid-beta ($\text{A}\beta$) peptide to neurological dysfunction in Alzheimer's disease (AD). Insoluble $\text{A}\beta$ plaques in the AD patient brain contain high concentrations of advanced glycation end-products (AGEs) as well as transition metal ions. This research elucidated the roles of $\text{A}\beta$, sugars, and Cu^{2+} in the oxidative stress mechanism of AD at the molecular level. Mass spectral (MS) analysis of the reactions of $\text{A}\beta$ with two representative sugars, ribose-5-phosphate (R5P) and methylglyoxal (MG), revealed Lys-16 and Arg-5 as the primary glycation sites. Quantitative analysis of superoxide ($\text{O}_2^{\bullet-}$) production by a cyt *c* assay showed that Lys-16 generated four times as much $\text{O}_2^{\bullet-}$ as Arg-5. Lys-16 and Arg-5 in $\text{A}\beta_{1-40}$ are both adjacent

to histidine residues, which are suggested to catalyze glycation. Additionally, Lys-16 is close to the central hydrophobic core (Leu-17 – Ala-21) and to His-13, both of which are known to lower the pK_a of the residue, leading to increased deprotonation of the amine and an enhanced glycation reactivity compared to Arg-5. Gel electrophoresis results indicated that all three components of AD plaques – $A\beta_{1-40}$, sugars, and Cu^{2+} – are necessary for DNA damage. It is concluded that the glycation of $A\beta_{1-40}$ with sugars generates significant amounts of $O_2^{\cdot-}$, owing to the rapid glycation of Lys-16 and Arg-5. In the presence of Cu^{2+} , $O_2^{\cdot-}$ converts to hydroxyl radical ($HO\cdot$), the source of oxidative stress in AD.

KEY WORDS

Alzheimer's disease, Amyloid beta, oxidative stress, copper, glycation

INTRODUCTION

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by loss of memory and cognitive decline [1–3]. The two main hallmarks of AD observed in the brain are senile plaques [4, 5] and neurofibrillary tangles [6]. The plaques are composed of aggregated amyloid beta ($A\beta$) peptides with high concentrations of co-localized transition metal ions such as $Cu^{2+/+}$ (approximately 0.4 mM), $Fe^{3+/2+}$ (approximately 1 mM), and Zn^{2+} (approximately 0.9 mM), [1, 4, 7–9] while neurofibrillary tangles consist of hyperphosphorylated tau proteins [6]. Cells around plaques and neurofibrillary tangle-bearing neurons show abnormally high amounts of oxidatively modified proteins, lipids and DNA [10–15].

Based on these observations of AD brains, there are currently three main hypotheses surrounding $A\beta$ and the pathogenesis of AD: the amyloid cascade hypothesis, the transition metal ion dyshomeostasis hypothesis, and the oxidative stress hypothesis [11]. The amyloid cascade hypothesis asserts that the aggregation of $A\beta$ peptides during the formation of senile plaques is the main source of neurotoxicity [16]. The metal ion dyshomeostasis hypothesis states that impaired metal homeostasis, particularly of Zn^{2+} , $Cu^{2+/+}$, and $Fe^{3+/2+}$, leads to $A\beta$ aggregation and cellular damage [8, 17]. The oxidative stress hypothesis claims that large amounts of reactive oxygen species (ROS) are generated

in the AD brain, which oxidize proteins, lipids and DNA in cells around plaques, and in neurofibrillary tangle-bearing neurons in the AD brain [6, 10, 11].

How are ROS generated? Multiple theories exist explaining the generation of reactive oxygen species. Some argue that H_2O_2 is spontaneously generated during the aggregation of $\text{A}\beta$ [18, 19]. Others believe that $\text{A}\beta$, the redox active metal ions $\text{Cu}^{2+/+}$ and $\text{Fe}^{3+/2+}$ - either bound or unbound to the $\text{A}\beta$ peptides -, and physiological reductants such as ascorbate, are involved in the generation of ROS [3, 7, 20, 21]. However, neither of these theories are completely satisfactory.

Another possible source of oxidative stress in AD is the glycation of $\text{A}\beta$ by sugars. Glycation, the non-enzymatic addition of a sugar to an amine or guanidine group of an amino acid, is an appealing mechanism because $\text{O}_2^{\bullet-}$, an ROS, is generated during the glycation process [22]. In the glycation model, there is no need for biological reductants like ascorbate or metal ions whether bound to $\text{A}\beta$ or not, to produce $\text{O}_2^{\bullet-}$. In addition, a large amount of $\text{A}\beta$ in the plaques of AD patients is found in the form of advanced glycation end-products (AGE's), which strongly suggests the concurrence of glycation with the development of AD [23]. Furthermore, the glycation mechanism is the only mechanism which may explain why diabetics develop AD at a higher rate than non-diabetic individuals [24]. A high concentration of un-metabolized glucose would glycate $\text{A}\beta$, producing large amounts of $\text{O}_2^{\bullet-}$. Furthermore, the extremely slow rate of glycation by glucose could explain why the oxidative damage seen in AD takes decades to mature.

Despite the likelihood of glycation to provide a mechanistic account of AD, there has been no detailed study of $\text{A}\beta$ glycation at the molecular level. It is not known which amino acid residues in $\text{A}\beta$ are glycation-reactive, if glycation of $\text{A}\beta$ generates $\text{O}_2^{\bullet-}$, or how transition metal ions contribute to oxidative stress.

Herein, we report that the Lys-16 residue of $\text{A}\beta_{1-40}$ is the major glycation site, followed by Arg-5. Glycation of $\text{A}\beta$ fragments that contain Lys-16 produced four times as much $\text{O}_2^{\bullet-}$ as those without this residue and with Arg-5 only. When Lys-16 was substituted with glycine in $\text{A}\beta_{1-40}\text{K16G}$, the production of $\text{O}_2^{\bullet-}$ was reduced by 80 % compared to wild-type $\text{A}\beta_{1-40}$. $\text{A}\beta$ fragments with neither Lys-16 nor Arg-5 failed to produce $\text{O}_2^{\bullet-}$.

Furthermore, A β ₁₋₄₀ produced significantly more O₂^{•-} than human serum albumin (HSA), a protein with 84 more potential glycation sites than A β ₁₋₄₀. All three components of AD plaques – A β ₁₋₄₀, sugars, and Cu²⁺ – were necessary for DNA damage. We conclude that the glycation of Lys-16 and Arg-5 in A β ₁₋₄₀ with reactive sugars produces O₂^{•-}, which subsequently forms damaging HO• in the presence of Cu²⁺ via the Fenton reaction [25, 26].

MATERIALS AND METHODS

Chemicals. A β ₁₋₉, A β ₁₋₁₂, A β ₁₁₋₂₂, A β ₁₇₋₄₀, and A β ₁₋₄₀ were purchased from AnaSpec, Inc. *N* α -acetyl-A β ₂₅₋₃₅ was purchased from Sigma Aldrich. A β ₁₋₁₈, A β ₁₄₋₃₈ and mutant A β ₁₋₄₀K16G were custom-ordered from GenScript. HSA, MG, and R5P were purchased from Sigma-Aldrich. pUC19 Vector from *E. coli* was purchased from New England BioLabs Inc. All other chemicals were purchased from Sigma Aldrich. All reagents were used without further purification. The water used was purified using a Hydro Picotech 2 purification system, unless otherwise specified. The pH was controlled using 10 mM sodium phosphate buffer at pH 7.4.

Amyloid-beta preparation. All A β peptides were purchased as lyophilized powder. Some were directly solubilized using 3 mM NaOH, while others were re-lyophilized using 100% hexafluoroisopropanol (HFIP) [27] prior to solubilization with NaOH. Both methods yielded satisfactory results. Based on the molecular weight and peptide content information available from the provider, each stock solution was prepared to yield a concentration of 0.5 mM or 1.0 mM in phosphate buffer at pH 7.4. LCMS-grade water was used to prepare the A β ₁₋₄₀ used for MS experiments. The concentration of A β ₁₋₄₀ calculated using its molecular mass was the same as the one obtained using $\epsilon = 1280 \text{ M}^{-1} \text{ cm}^{-1}$ at 275 nm arising from Tyr-10 [28]. The concentrations of other fragments were calculated due to the lack of information on their molar absorptivity values. The peptides were used as soon as they were prepared.

Sugars and amino acids stock solutions. Stock solutions of ribose-5-phosphate monosodium, MG, *N* α -acetyl-lysine (*N* α -Ac-Lys) and *N* α -acetyl-arginine (*N* α -Ac-Arg) were prepared in 10 mM phosphate buffer (pH 7.4). An additional stock solution of R5P

was prepared in LCMS-grade water for A β ₁₋₄₀ MS glycation experiments. The pH of all stock solutions was adjusted using NaOH and HCl. The phosphate buffer used to create all stock solutions was chelated using Chelex100 to avoid metal contamination. A stock solution of CuCl₂ was prepared in a water medium.

Electrospray Ionization Liquid-Chromatography Mass Spectrometry (ESI LCMS). Liquid Chromatography coupled to a Mass Spectrometer was used to identify the reaction products of each system. The mass spectrometry analyses of all peptides and the tandem mass spectrometry analyses of the A β fragments were performed using a 1100 Series LC, coupled to an LC-MSD Trap XCT Plus system from Agilent Technologies. An Eclipse X-D8-C8 5 μ m 4.6x150 mm column was used for separation, and the liquid chromatography unit was equipped with a photo diode array detector. Solvent-gradient elution was performed with (A) LCMS-grade water + 0.1 % trifluoroacetic acid (TFA) and (B) LCMS-grade acetonitrile + 0.1 % TFA solvent at a flow rate of 0.5 mL/min. The gradient started at 0 % B, increased to 70 % B after 30 minutes, and then maintained at 100 % B for ten minutes. Chromatography of all other molecules was achieved by isocratic elution using 95 % A and 5 % B. The mass spectrometer was set to scan in the range of 50–2200 m/z , detecting positive ions with electrospray ionization mass spectrometry (ESI–MS). The nitrogen nebulizer pressure was set to 50 psi, the dry gas (nitrogen) flow to 9 L/min, and the dry temperature to 365 °C. The ion trap was set to collect targeted molecules and acquire their fragmentation mass spectra. Experimental fragmentation spectra for glycated A β ₁₋₉ and A β ₁₁₋₂₂ were manually compared to a list of calculated m/z values for possible *b*, *a*, and *y* ions (Chart 2).

Quadrupole Time of Flight Electrospray Ionization Liquid Chromatography Mass Spectrometry (QTOF-ESI-LCMS). Analysis was performed using a UPLC Waters Acquity I Class LC, coupled to a Waters Xevo G2-XS Quadrupole Time of Flight mass spectrometer. An Acquity UPLC CSH C18 1.7 μ m 2.1x100 mm column was used to perform the chromatographic separation. The mass spectrometer was set to detect and fragment positive ions in ES⁺ Sensitivity MSe mode with 6 eV collision energy (function 1). A gradient collision energy from a low 15 eV to a high 25 eV function was used to maximize the number of fragments (function 2). A LockSpray function was employed

using a solution of Leucine Enkephalin (200 pg/ μ L) to correct for deviations of mass during analysis. The mass spectrometer was calibrated using 0.5 mM sodium formate. Solvent-gradient elution was performed using (A) LC-MS grade water + 0.1 % formic acid and (B) Optima-grade acetonitrile at a flow rate of 0.1 mL/min. The elution gradient started at 100 % A and changed to 30 % A and 70 % B in the first 20 minutes. Solvent A was decreased to 0 % at 20.67 minutes and maintained at 0 % for ten minutes. The fragmentation data was analyzed using the Waters UNIFI Scientific Information System.

Cytochrome c assay. The peptide solutions were combined with appropriate volumes of MG and the reactions were corrected to their final concentration using 10 mM phosphate buffer at pH 7.4. All reaction mixtures were incubated for 5 hours at 37 °C. A 150 μ L aliquot of each reaction was combined with 150 μ L of 40 μ M cytochrome c solution (prepared in water). All spectra were obtained immediately after addition of cytochrome c. UV/Vis spectra were obtained every 5 minutes over a 60-minute period at 37 °C in 1 cm path length micro quartz cells employing a Varian Cary 4000 Spectrophotometer with Cary Win UV kinetic assay software. All kinetic experiments were performed in triplicate.

Agarose gel electrophoresis DNA cleavage assay. All reactions were incubated for 5 hours in the absence or presence of 0.01 mM CuCl_2 at 37 °C. A 16 μ L aliquot of each reaction was combined with 4 μ L of 50 ng/ μ L solution of pUC19 vector (previously prepared in sterile water). The new reaction mixture was incubated at 37 °C for 1 hour. Each 20 μ L sample was combined with 5 μ L of 5X agarose gel loading dye and loaded onto a 0.7 % agarose gel containing 0.5 % Ethidium Bromide. A 1 kb DNA ladder from New England BioLabs was used as a reference. DNA separation was achieved at 150 V over a 45-minute period. All analyses were performed using a BioRad Molecular Imager ChemiDocTM XRS+ equipped with Image LabTM 5.1 Software.

High-Performance Liquid Chromatography (HPLC). HPLC chromatograms were obtained using a Waters 2695 Alliance liquid chromatograph equipped with a Waters 2996 photodiode array detector (DAD). A Waters Atlantis dC18 5 μ m 4.6x250 mm column was used to perform separation. Isocratic elution was performed with (A) HPLC-grade water and (B) HPLC-grade acetonitrile at 25 °C. The solvent mixture was set to 99 % (A) and 1 % (B). Data acquisition was performed using the Waters Empower software.

RESULTS

The two representative sugars employed in this study were methylglyoxal (MG) and ribose-5-phosphate (R5P) (Chart 1).

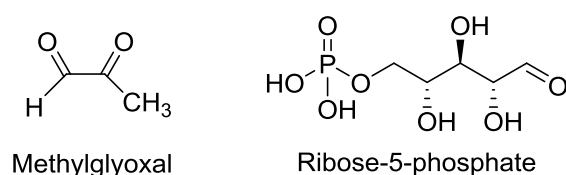


Chart 1. Structures of methylglyoxal (MG) and ribose-5-phosphate (R5P).

MG, a highly-reactive dicarbonyl sugar, is produced *in vivo* by the spontaneous dephosphorylation of glycolytic intermediates, by the autoxidation of glucose, and by post-Amadori glycation reactions. As an acyclic molecule with two open carbonyl groups, MG rapidly reacts with amines to generate a variety of AGEs, as well as superoxide radical anion, $O_2^{\bullet-}$ [29]. The dicarbonyl moiety provides MG the ability to form cross-links, allowing the spontaneous formation of intra- and intermolecular covalent linkages between peptide residues [30].

R5P, in contrast, exists in solution primarily (> 99 %) as a cyclic hemiacetal that is unreactive to amine attack. The small amount of available acyclic R5P, however, is glycation-reactive, leading to an Amadori adduct, and subsequently to other AGEs including MG.

The excessive cross-linking that results from glycation of MG was observed to preclude effective DNA separation in agarose gel electrophoresis experiments, and to hinder the identification of modified monomeric peptides in MS experiments. On the contrary, glycation by R5P appeared to be sufficiently slow to minimize cross-linking while still providing sufficient DNA damage and effective DNA separation, as well as allowing detection of monomeric peptides in MS analysis.

1. R5P glycates at Lys-16 in A β ₁₋₄₀. Figure 1 shows the mass spectrum of the reaction between A β ₁₋₄₀ and R5P. The peaks at $m/z = 1444.1$ and 1514.6 are due to the $[M+3H]^{3+}$ and $[M+R5P+3H]^{3+}$ ions of free and modified A β ₁₋₄₀, respectively. The

difference of $211.5 \text{ g}\cdot\text{mol}^{-1}$ in molecular weight between the two peaks is the molecular weight of a single R5P addition (minus H_2O), which indicates that among the four possible glycation sites in $\text{A}\beta$ (N-terminus, Arg-5, Lys-16, and Lys-28), only one of them is glycation-reactive under these reaction conditions.

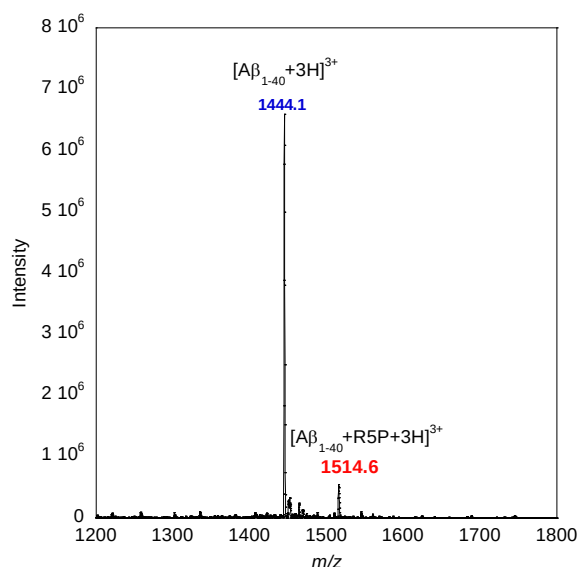


Figure 1. Mass spectrum of the reaction between $\text{A}\beta_{1-40}$ (0.01 mM) and R5P (10 mM). 6 h incubation at 50°C in LCMS-grade water.

The $[\text{M}+\text{R5P}+3\text{H}]^{3+}$ MS peak was further analyzed by UPLC-ESI-QTOF- MS^E . The QTOF- MS^E spectrum shows daughter ion fragments generated by collision induced dissociation (CID). They are designated as *b*, *y*, and *a* ions (Chart 2) [31].

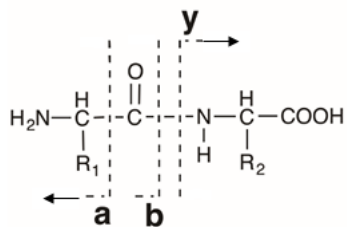


Chart 2. Daughter ion nomenclature of peptide fragmentation.

The QTOF-MS^E spectrum of the product formed in the reaction between A β ₁₋₄₀ and R5P (Figure 2) shows 15 unmodified peaks and two modified peaks. The fragmentation of the A β ₁₋₄₀-R5P Amadori adduct corresponding with the peaks in the QTOF-MS^E spectrum are shown in Chart 3.

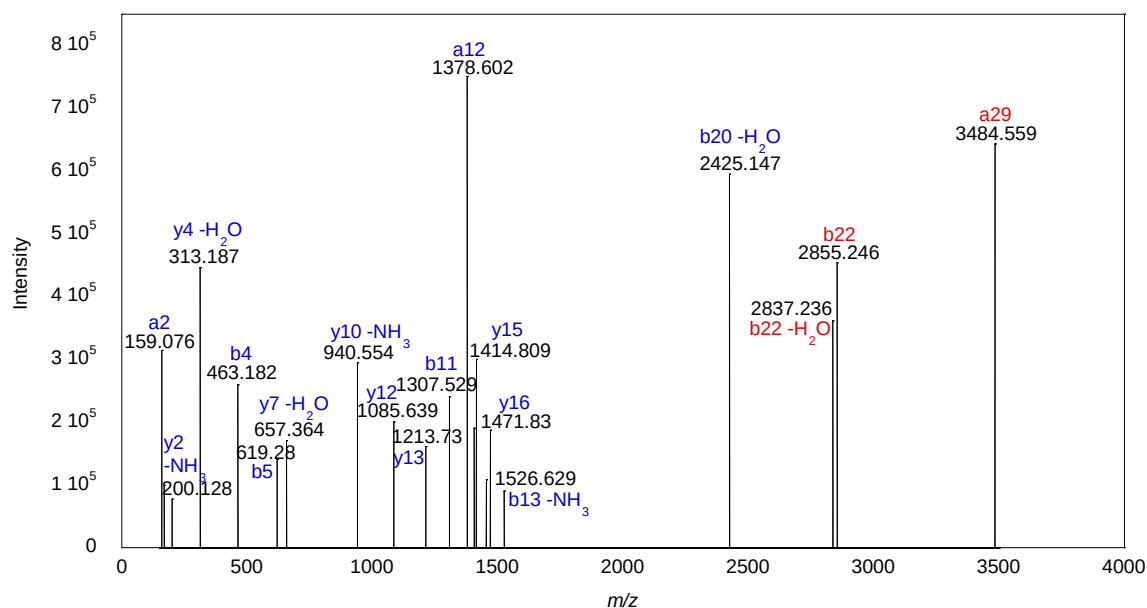


Figure 2. QTOF-MS^E spectrum of the product formed in the reaction between A β ₁₋₄₀ (0.01 mM) and R5P (10 mM). Unmodified and modified peaks are designated in blue and red color, respectively.

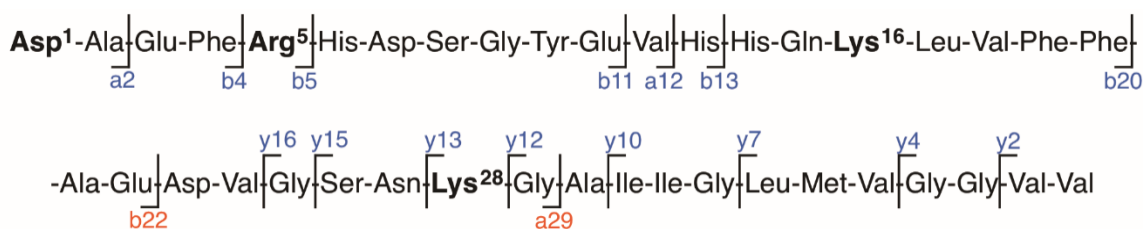


Chart 3. Schematic representation of the fragmentation obtained from the product of A β ₁₋₄₀ with R5P.

Fragments a2, b4, b5, b11, a12 and b13 are not modified, indicating that neither the N-terminus nor Arg-5 reacted with R5P. Fragments y13, y15, and y16 are not modified, indicating that Lys-28 is not the glycation site. This suggests Lys-16 as the only glycation

site candidate. The two modified peaks, b22 and a29, include Lys-16 and confirm that Lys-16 is the glycation site.

The existence of the unmodified peak b20 is puzzling because it contains Lys-16 and should have been modified. However, observing two fragments with modified and unmodified Lys-16, as well as free peptide in the MS/MS spectrum of the $[A\beta_{11-22}+R5P+H]^+$ ion (Figure S1) suggests that the bond between Lys-16 and R5P can be broken during the fragmentation process. Our confidence that Lys-16 is the major glycation site is further elevated by the observation that any A β fragment that includes Lys-16 generates far more $O_2^{\bullet-}$ compared to those without Lys-16 (*vide infra*). The inability of Arg-5 and Lys-28 to react with R5P was further confirmed by MS/MS analysis of $A\beta_{1-9}/R5P$ reaction (Figure S2), and by MS analysis of the $A\beta_{25-35}/R5P$ and *N* α -acetyl- $A\beta_{25-35}/R5P$ reactions (Figure S3).

2. MG glycates at Arg-5 in A β_{1-9} . Figure 3 displays (a) the mass spectrum of the reaction of MG with A β_{1-9} and (b) the MS/MS spectrum of the glycation reaction product. Mass spectrum (a) shows four major peaks with $m/z = 517.0$, 553.2, 1033.0, and 1105.1. The peaks with $m/z = 517.0$ and 1033.0 are attributed to the $[M+2H]^{2+}$ and $[M+H]^+$ ions of free A β_{1-9} , respectively. The peaks with $m/z = 553.2$ and 1105.1 both display a mass increase of $72 \text{ g}\cdot\text{mol}^{-1}$ and are attributed to the $[M+MG+2H]^{2+}$ and $[M+MG+H]^+$ ions of the same peptide modification. A change in mass of $72 \text{ g}\cdot\text{mol}^{-1}$ is consistent with the formation of the well-characterized dihydroxyimidazolidine modification on Arg-5.[32]

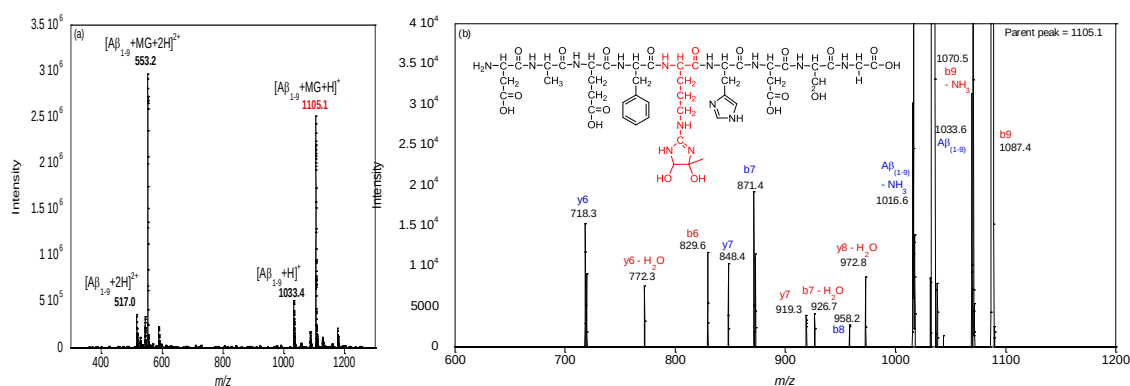


Figure 3. (a) MS and (b) MS/MS spectra of the product formed in the reaction between MG (30 mM) and A β_{1-9} (0.3 mM). Incubation: 2 h at 37 °C, pH 7.4. Modified peaks are red and unmodified peaks are blue.

The MS/MS spectrum shows seven modified peaks and six unmodified peaks. The peaks with $m/z = 1033.6$ and 1016.6 in the tandem mass spectrum of the *modified* parent peak ($m/z = 1105.1$) correspond to free A β_{1-9} and they indicate that the modification on A β_{1-9} can be fully cleaved during fragmentation, giving rise to a combination of modified and unmodified peaks. The two potentially reactive sites on this peptide are the *N*-terminus and Arg-5. The unmodified peaks provide no insight into the modification of the peptide. The modified peaks y6, y7, and y8 contain Arg-5 (Chart 4) but not the *N*-terminus, and therefore, their presence indicates that Arg-5 must have been the amino acid residue that was modified upon reaction of MG with A β_{1-9} .

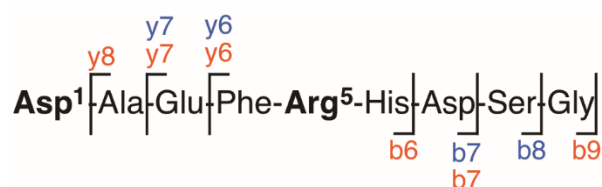


Chart 4. Schematic representation of the fragmentation obtained from the product of A β_{1-9} with MG.

3. MG glycate at Lys-16 in A β_{11-22} . Figure 4 shows the (a) mass spectrum of the reaction of MG with A β_{11-22} and (b) the MS/MS spectrum of their reaction product. Mass spectrum (a) shows two peaks with $m/z = 1483.9$ and 742.6 . These peaks correspond to the $[M+H]^+$ and $[M+2H]^{2+}$ ions of free A β_{11-22} , respectively. The peaks with $m/z = 1610.1$ and 805.8 both show an increase in the mass of the peptide by $126.2 \text{ g} \cdot \text{mol}^{-1}$. These m/z values are attributed to the $[M+MG+H]^+$ and $[M+MG+2H]^{2+}$ ions of the modified peptide, and are consistent with the formation of two modifications: *N* ϵ -carboxy-ethyl-lysine (CEL) [22, 33] on the Lys residue, and a Schiff base on the *N*-terminus.

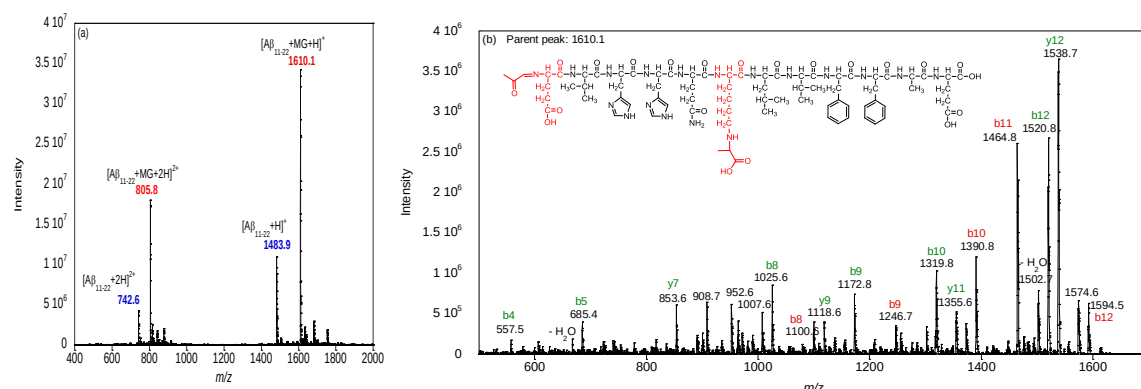


Figure 4. (a) MS and (b) MS/MS spectra of product formed in the reaction between MG (30 mM) and A β ₁₁₋₂₂ (0.3 mM). Incubation: 2 h at 37 °C, pH 7.4. Peaks with both modifications are red and with only the N-terminal modification are green.

The tandem mass spectrum (Figure 4(b)) displays seven peaks that are modified by 126 g·mol⁻¹ and thus contain both modifications (red), and eight peaks modified by 54 g·mol⁻¹ that appear to contain only the *N*-terminal modification (green).

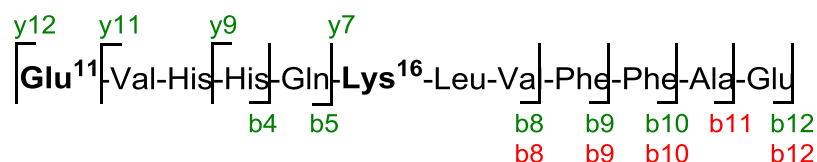


Chart 5. Schematic representation of the fragmentation obtained from the product of A β_{11-22} with MG.

The presence of dual fragments b8, b9, b10, and b12 (Chart 5) that display a mass increase of both $54 \text{ g}\cdot\text{mol}^{-1}$ and $126 \text{ g}\cdot\text{mol}^{-1}$, suggests that, as observed on other MS/MS spectra (*vide supra*), the fragmentation process appears to cleave the Lys-16 modification. The only two possible sites of reaction on this peptide fragment are the *N*-terminus and Lys-16. Fragments b4 and b5 contain only the *N*-terminus and are modified by $54 \text{ g}\cdot\text{mol}^{-1}$ of the total $126 \text{ g}\cdot\text{mol}^{-1}$. This change in mass is consistent with the formation of a Schiff base on the *N*-terminal amine. Fragments b8, b9, b10, b11, and b12 contain both the *N*-terminus and Lys-16 and are modified by $126 \text{ g}\cdot\text{mol}^{-1}$. Subtracting $54 \text{ g}\cdot\text{mol}^{-1}$ due to the *N*-terminal Schiff base, the remaining modification must be $72 \text{ g}\cdot\text{mol}^{-1}$.

This change in mass is consistent with the formation of the well characterized N_ϵ -carboxy-ethyl-lysine (CEL) modification [33]. The N -terminal modification of this fragment is inconsequential, however, because this particular amine functionality is unavailable for reaction in $A\beta_{1-40}$, as it is part of the peptide backbone.

4. Glycation of MG with $A\beta$ generates $O_2^{\bullet-}$ that reduces $cyt\ c^{3+}$ to $cyt\ c^{2+}$. The concentration of $O_2^{\bullet-}$ produced by glycation reactions was indirectly measured by a cytochrome c^{3+} ($cyt\ c^{3+}$) reduction assay. Ferri-cytochrome c , $cyt\ c^{3+}$, is known to be reduced to ferrous-cytochrome c , $cyt\ c^{2+}$, by $O_2^{\bullet-}$ [34].

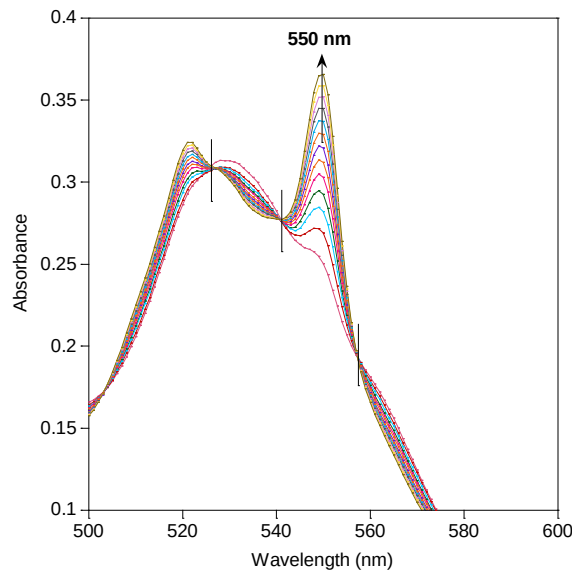


Figure 5. Scanning kinetic spectra as $cyt\ c^{3+}$ is reduced to $cyt\ c^{2+}$ by $O_2^{\bullet-}$.

Figure 5 depicts the scanning kinetic spectra of $cyt\ c^{3+}$ reduction to $cyt\ c^{2+}$ by a glycation reaction over time, showing an increase in absorbance at 550 nm due to the generation of $cyt\ c^{2+}$. From Beer's law in a 1 cm cell,

$$A_{550} = \epsilon_{550}^{3+}[cyt\ c^{3+}] + \epsilon_{550}^{2+}[cyt\ c^{2+}] = \epsilon_{550}^{3+}([cyt\ c^{3+}]_o - [cyt\ c^{2+}]) + \epsilon_{550}^{2+}[cyt\ c^{2+}],$$

the concentration of $\text{cyt } c^{2+}$ can be obtained using the calculated value of $\epsilon_{550}^{3+} = 9.15 \times 10^3 \text{ M}^{-1} \text{ cm}^{-1}$, and $\epsilon_{550}^{2+} = 2.77 \times 10^4 \text{ M}^{-1} \text{ cm}^{-1}$ from the literature [35].

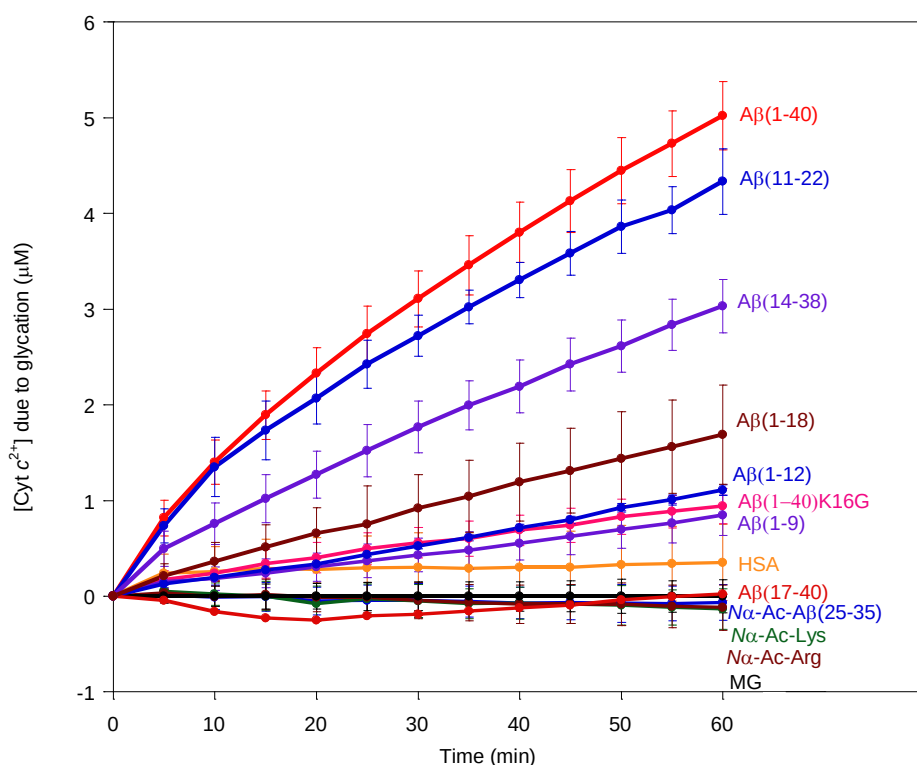


Figure 6. Concentration of $\text{cyt } c^{2+}$ vs time from the reactions of MG (1 mM) with Aβ₁₋₄₀ (0.1 mM), Aβ₁₁₋₂₂ (0.1 mM), Aβ₁₄₋₃₈ (0.1 mM), Aβ₁₋₁₈ (0.1 mM), HSA (4.5 μM), Aβ₁₋₁₂ (0.1 mM), Aβ₁₋₄₀K16G (0.1 mM), Aβ₁₋₉ (0.1 mM), Aβ₁₇₋₄₀ (0.1 mM), Nα-acetyl-Aβ₂₅₋₃₅ (0.1 mM), Nα-acetyl-Arg (0.1 mM), Nα-acetyl-Lys (0.1 mM). [cyt c^{3+}] = 20 μM. The reactions were pre-incubated for 5 h at 37 °C in 10 mM phosphate buffer (pH 7.4) before $\text{cyt } c^{3+}$ was added. n = 3.

Figure 6 displays the concentrations of $\text{cyt } c^{2+}$ generated from the reactions of MG with various Aβ fragments, HSA, Lys, and Arg. A standard solution of MG was used as a control to account for the generation of $\text{O}_2^{\bullet-}$ by autoxidation. Under our conditions, MG autoxidation generated 1 μM $\text{cyt } c^{2+}$ after a 5 h pre-incubation, which was subtracted from the total concentration of $\text{cyt } c^{2+}$ generated in other systems.

Glycation of Aβ₁₋₄₀ produced the largest concentration of $\text{cyt } c^{2+}$, 5 μM, at the end of a 60-min period. Fragments containing only Lys-16, Aβ₁₁₋₂₂ and Aβ₁₄₋₃₈, produced approximately 4 μM and approximately 3 μM of $\text{cyt } c^{2+}$, respectively. Glycation of the

fragments containing only Arg-5, A β ₁₋₁₂, A β ₁₋₄₀K16G, and A β ₁₋₉, each produced approximately 1 μ M cyt c^{2+} . A fragment with both Lys-16 and Arg-5, A β ₁₋₁₈, generated approximately 1.7 μ M cyt c^{2+} . Reaction of the fragments A β ₁₇₋₄₀ and N_{α} -Ac-A β ₂₅₋₃₅, which contain neither Arg-5 nor Lys-16, did not generate any cyt c^{2+} . N_{α} -Ac-Lys and N_{α} -Ac-Arg also did not produce any cyt c^{2+} . Although HSA (4.5 μ M) has 60 Lys (2.7 mM) and 27 Arg (1.2 mM) residues, it produced only approximately 0.4 μ M cyt c^{2+} . These results are summarized in Table 1.

Table 1. Summary of cyt c^{2+} generated in a 60-min period from the reaction of MG (1 mM) with A β peptides (0.1 mM), Lys (0.1 mM), Arg (0.1 mM), or HSA (4.5 μ M).

A β Peptides, Lys, Arg, or HSA	[Cyt c^{2+}] (μ M)
A β ₁₋₄₀	5.0 $\pm$ 0.4
A β ₁₁₋₂₂	4.3 $\pm$ 0.3
A β ₁₄₋₃₈	3.0 $\pm$ 0.3
A β ₁₋₁₂	1.1 $\pm$ 0.3
A β ₁₋₄₀ K16G	0.9 $\pm$ 0.2
A β ₁₋₉	0.8 $\pm$ 0.2
A β ₁₋₁₈	1.7 $\pm$ 0.5
A β ₁₇₋₄₀	0.0 $\pm$ 0.3
N_{α} -Ac-A β ₂₅₋₃₅	-0.1 $\pm$ 0.2
N_{α} -Ac-Lys	-0.1 $\pm$ 0.2
N_{α} -Ac-Arg	-0.1 $\pm$ 0.2
HSA	0.4 $\pm$ 0.3

5. Glycation and Cu^{2+} are necessary to damage DNA. Figure 7 shows the time-lapse agarose gel electrophoresis DNA cleavage assay of reactions containing R5P and Cu^{2+} with and without $\text{A}\beta_{1-40}$. Compared to the DNA ladder in Lane 1, Lanes 2 through 9 display DNA bands at 1.5 kB (supercoiled), approximately 1.8 kB (linear), and 3 kB (nicked). At $t = 0$, both reactions show a high-intensity band of supercoiled DNA and a low-intensity band of nicked DNA (Lanes 2 and 6). As the reaction time progresses, linear DNA bands appear for both reactions. However, the most dramatic change in band intensities is seen for the decrease of supercoiled DNA bands of R5P/ Cu^{2+} and R5P/ Cu^{2+} / $\text{A}\beta_{1-40}$ reactions at $t = 3$ h; the latter (Lane 9) is much weaker than the former (Lane 5). This indicates that the reaction of R5P/ Cu^{2+} / $\text{A}\beta_{1-40}$ generated a larger amount of $\text{HO}\cdot$ to damage DNA than the reaction of R5P/ Cu^{2+} . Glycation of R5P with $\text{A}\beta_{1-40}$ generated $\text{O}_2^{\cdot-}$ (Figure S4) that subsequently converted to DNA-damaging $\text{HO}\cdot$ in the presence of Cu^{2+} .

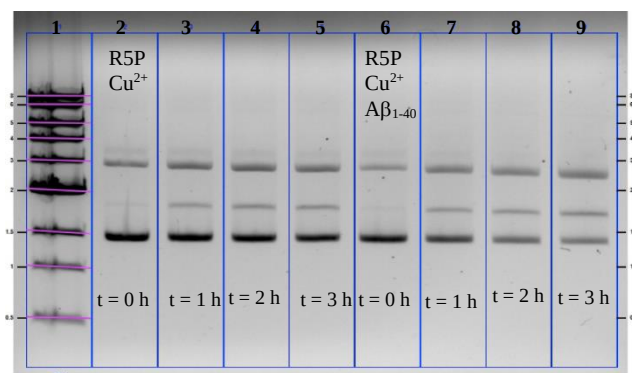


Figure 7. Agarose gel electrophoresis (0.7 %) DNA cleavage assay of the reactions of R5P/ Cu^{2+} and R5P/ Cu^{2+} / $\text{A}\beta_{1-40}$. [R5P] = 0.5 mM, [$\text{A}\beta$] = 0.1 mM, [CuCl_2] = 10 μM , [pUC19] = 10 ng/ μL . Stain: Ethidium Bromide. Voltage: 150 V. Incubation: 5 hours at 37 $^\circ\text{C}$, pH 7.4.

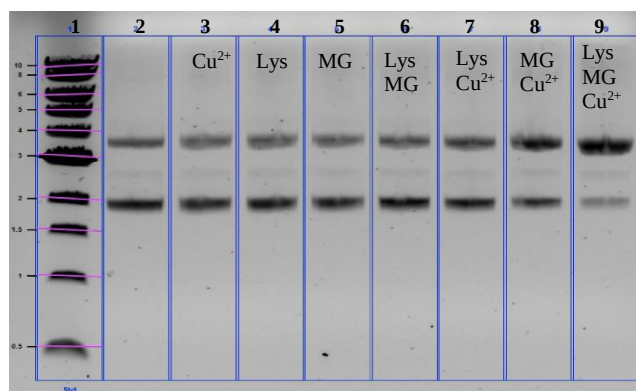


Figure 8. Agarose gel electrophoresis (0.7 %) DNA cleavage assay: [Cu^{2+}] = 0.4 mM, [Lys] = 4 mM, [MG] = 4 mM, [pUC19] = 10 ng/ μL , Stain: Ethidium Bromide. Voltage: 150 V. Incubation: 5 hours at 37 $^\circ\text{C}$, pH 7.4.

In order to further investigate the required components for DNA damage, a series of reactions containing different combinations of Lys, MG, and Cu^{2+} were analyzed by gel electrophoresis (Figure 8). Lys and MG were used to model $\text{A}\beta_{1-40}$ and a sugar, respectively. As shown in Figure 8, solutions containing Cu^{2+} alone (Lane 3), Lys alone (Lane 4), MG alone (Lane 5), Lys/MG (Lane 6), and Lys/ Cu^{2+} (Lane 7) were unable to induce DNA damage. The solution of MG/ Cu^{2+} (Lane 8) shows a slight increase of nicked DNA and decrease in supercoiled DNA, but Lys/MG/ Cu^{2+} shows significant damage. The results confirm that three components are necessary to produce a significant amount of $\text{HO}\cdot$.

6. Glycation in the presence of Cu^{2+} oxidizes 5'-dGMP to 8-oxo-5'-dGMP. The reactions of Lys and R5P with and without Cu^{2+} were treated with 5'-dGMP to elucidate the origin of DNA damage. The reaction with Cu^{2+} showed an HPLC peak whose UV spectrum matches that of 8-oxo-5'-dGMP (Figure S5(a)).[36] The production of 8-oxo-5'-dGMP was further confirmed by MS analysis showing an 8-oxo-5'-dGMP peak at $m/z = 364.2$ (Figure S5(b)). The reaction without Cu^{2+} showed no 8-oxo-5'-dGMP peak in either HPLC or MS analysis (data not shown). The results indicate that $\text{HO}\cdot$ was produced from the reaction mixture of Lys/R5P/ Cu^{2+} and oxidized 5'-dGMP to 8-oxo-5'-dGMP.

DISCUSSION

The MS analyses revealed that R5P glyicates only at Lys-16, and that MG glyicates at both Lys-16 and Arg-5. The reaction of arginine residues with MG has been well defined [37]. The equilibrium constant for an imidazolone modification on an Arg is approximately 500-fold greater than the attachment of MG onto a Lys [37]. While sugars (such as R5P) are known to generate dicarbonyl products (such as 3-deoxyglucosone), imidazolone modifications of Arg residues by sugars are not typically observed in physiological conditions [33]. Carboxy ethyl lysine modifications of Lys residues by MG, such as the type observed here, have been previously defined [33]. No glycation products were observed at the *N*-terminus or Lys-28 during reactions with R5P or MG.

The cyt *c* assays were consistent with the MS analyses. Fragments without Arg-5 and Lys-16, but with Lys-28 (A β ₁₇₋₄₀ and N α -Ac-A β ₂₅₋₃₅) did not generate significant O₂^{•-} to reduce cyt *c*³⁺, confirming that Lys-28 is not the glycation site. The fragments lacking Lys-16 but containing Arg-5 (A β ₁₋₉, A β ₁₋₁₂, and A β ₁₋₄₀K16G) generated approximately 1 μ M cyt *c*²⁺. Fragments containing Lys-16 without Arg-5 (A β ₁₁₋₂₂) generated approximately 4 μ M cyt *c*²⁺. This means the approximately 5 μ M cyt *c*²⁺ produced by A β ₁₋₄₀ consists of approximately 1 μ M cyt *c*²⁺ by Arg-5 and approximately 4 μ M cyt *c*²⁺ by Lys-16. However, fragments that lack key residues around Lys-16 produced significantly less cyt *c*²⁺. A β ₁₄₋₃₈ produced approximately 3 μ M, and A β ₁₋₁₈ generated a total of approximately 1.7 μ M cyt *c*²⁺, approximately 1 μ M by Arg-5 and approximately 0.6 μ M by Lys-16.

Why does Lys-16 glycate four times faster than Arg-5? Why do fragments containing Arg-5 glycate more actively than N α -Ac-Arg? Why is no observable glycation seen on Lys-28? Why is there variability seen in the glycation ability of Lys-16 for the various fragments? There seems to be a site-specificity for glycation.

An amine must be in its neutral form (-NH₂) to perform the nucleophilic attack required to initiate glycation [34]. At neutral pH, the amino group on the side chain of monomeric Arg and Lys primarily exists in their cationic form (-NH₃⁺) with a pK_a of approximately 12,[38] and 10.53, [39] respectively. However, in a protein environment, the pK_a of their amine and guanidine side chains can be lowered. In a hydrophobic environment such as the interior of a protein, the pK_a of Lys could be decreased to as low as 5.3 [39]. While not buried inside a large protein, Lys-16 is adjacent to a region known as the ‘central hydrophobic core’ of A β , a group of hydrophobic residues denoted A β ₁₇₋₂₁ (Leu-17 – Ala-21) in the middle of the A β sequence [40]. The proximity of Lys-16 to this region of the peptide may have an influence on the side chain’s pK_a, favoring its neutral form over the charged form. Zhao *et al.* predict the pK_a of Lys-16 to be 9.7 in an aqueous environment, 0.8 pK_a units lower than the pK_a of the amino acid [41].

Comparing three fragments with Lys-16, A β ₁₁₋₂₂, A β ₁₄₋₃₈ and A β ₁₋₁₈, proves the importance of the adjacent hydrophobic core (Leu-17 – Ala-21) for glycation. The former two fragments with adjacent hydrophobic core produced approximately 4 μ M and

approximately 3 μM cyt^{2+} , respectively. However, $\text{A}\beta_{1-18}$ that lacks the hydrophobic core generated only approximately 1.7 μM cyt^{2+} even with its contribution from Arg-5.

Another factor that favors the glycation of the Lys-16 amine side chain can be its proximity to His-13 (Figure 9). Histidine residues are known to be involved in deprotonating adjacent residues via H-Bonding, lowering the pK_a of amine groups [42]. Indeed, in the fragment $\text{A}\beta_{14-38}$, cyt^{2+} production (approximately 3 μM) drops by 25% of $\text{A}\beta_{11-22}$ (approximately 4 μM). We attribute this drop to the removal of His-13 in $\text{A}\beta_{14-38}$.

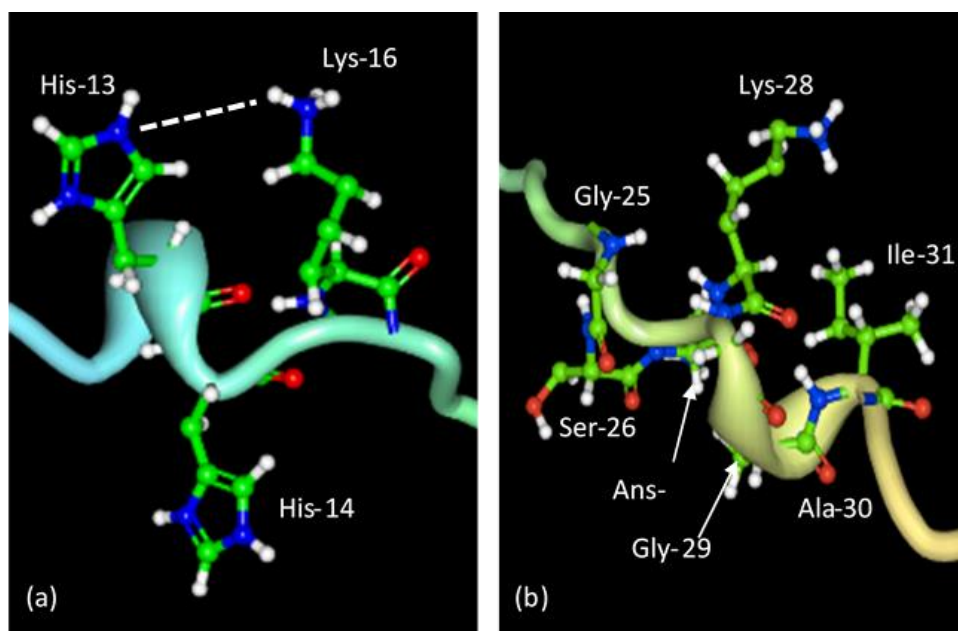


Figure 9. Orientation of (a) Lys-16 and (b) Lys-28 in $\text{A}\beta$ with their neighboring amino acid residues. Source: Protein workshop 1Z0Q by Protein Data Bank [43].

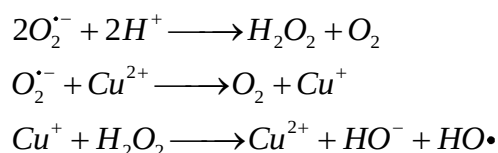
Although the guanidine functional group of Arg-5 may not be as deprotonated as that of Lys-16, it still showed higher propensity for glycation compared to its *N*-acetylated amino acid form. The pK_a of the side chain of Arg is reported to reach 13 in a protein environment [38]. However, the adjacent His-6 may lower the pK_a of Arg-5 and/or may serve as an acid/base catalyst in the rate-determining Amadori rearrangement, thus driving glycation [42]. Although we did not examine the role of His-14 in the glycation ability of Lys-16, we predict the removal of His-14 would decrease $\text{O}_2^{\cdot-}$ production of Lys-16.

The side-chain of Lys-28 is farther away from the central hydrophobic core (Figure 9). With no amino acid nearby to lower the pK_a of its amine group or His residues to catalyze an Amadori rearrangement, Lys-28 does not glycate. It remains identical to N_α -Ac-Lys.

The site-specificity nature of glycation is further strengthened when the reaction is compared to the glycation of HSA. It is striking to see that $A\beta_{1-40}$ generates approximately 12 times more $\text{cyt } c^{2+}$ (5 μM) than HSA (0.4 μM) at the same concentration of available glycation sites. The 0.1 mM $A\beta_{1-40}$ has a total of 0.4 mM glycation sites (0.2 mM Lys from two Lys residues, 0.1 mM Arg from one Arg residue and 0.1 mM glycation site from the N -terminus), while the 4.5 μM HSA has a total of 0.4 mM glycation sites (0.27 mM Lys from 60 Lys residues, 0.12 mM Arg from 27 Arg residues, and 0.04 mM from the N -terminus). The glycation sites of HSA must not be adjacent to strong glycation promoters such as hydrophobic regions of His residues.

Recently, Lim's group synthesized a metal complex, $\text{Co(II)(tetra-N-methylated cyclam)}$, that can hydrolyze $A\beta$ to small aggregation resistant fragments such as $A\beta_{1-12}$, $A\beta_{1-18}$, and $A\beta_{14-38}$ [44]. However, our results indicate that this fragmentation pattern would not prevent oxidative stress. All three fragments, $A\beta_{1-12}$, $A\beta_{1-18}$, and $A\beta_{14-38}$, produced $\text{O}_2^{\bullet-}$, with 20 %, 32 %, and 60 %, respectively, of the $\text{O}_2^{\bullet-}$ generated by the glycation of full-length $A\beta_{1-40}$. A molecule that can hydrolyze $A\beta$ into five fragments, $A\beta_{1-4}$, Arg-5, $A\beta_{6-15}$, Lys-16, and $A\beta_{17-40}$ may be developed as a superior AD drug, as none of these fragments would contribute to the oxidative stress of AD.

Although $A\beta_{1-40}$ has a superior ability to glycate and produce a significantly large amount of $\text{O}_2^{\bullet-}$, it requires Cu^{2+} (or Fe^{3+}) to cause DNA damage. As shown in Figure 7, significant DNA damage was only observed when all three components, $A\beta_{1-40}$, R5P and Cu^{2+} , were in the system. The importance of glycation and Cu^{2+} in DNA damage and in the oxidation of guanine were further shown in Figures 8 and 9. Glycation between amino acids and sugars generated $\text{O}_2^{\bullet-}$, which converts to $\text{HO}\bullet$ via the Fenton reaction [26].



Thus, the utmost important role of A β in the oxidative stress mechanism of AD appears to be its ability to produce a significant amount of $O_2^{\cdot-}$ through glycation, which brings about DNA damage in the presence of Cu^{2+} . Our work may give molecular insights into many *in vivo* and *in vitro* experimental results that claim A β exacerbates DNA damage [4, 17].

ACKNOWLEDGEMENTS

The authors thank Middlebury College, and the Arnold and Marble Beckman Foundation for their financial support for this work. We also acknowledge Ms. Jody Smith and Dr. Royston Quintyn for their help with mass spectrometry and MS data analysis.

REFERENCES

1. Rauk A (2009) The chemistry of Alzheimer's disease. *Chem Soc Rev* 38:2698–2715. doi: 10.1039/b807980n
2. (2013) Latest Alzheimer's Facts and Figures. In: Latest Facts Fig. Rep. Alzheimers Assoc. <http://www.alz.org/facts/overview.asp>. Accessed 19 Nov 2016
3. Telpoukhovskaia MA, Orvig C (2013) Werner coordination chemistry and neurodegeneration. *Chem Soc Rev* 42:1836–1846. doi: 10.1039/c2cs35236b
4. Li X-H, Du L-L, Cheng X-S, et al (2013) Glycation exacerbates the neuronal toxicity of β -amyloid. *Cell Death Dis* 4:e673. doi: 10.1038/cddis.2013.180
5. Angeloni C, Zamboni L, Hrelia S (2014) Role of methylglyoxal in Alzheimer's disease. *BioMed Res Int*. doi: 10.1155/2014/238485
6. Brion JP (1998) Neurofibrillary tangles and Alzheimer's disease. *Eur Neurol* 40:130–140.
7. Jiang D, Li X, Williams R, et al (2009) Ternary complexes of iron, amyloid-beta, and nitrilotriacetic acid: binding affinities, redox properties, and relevance to iron-induced oxidative stress in Alzheimer's disease. *Biochemistry (Mosc)* 48:7939–7947. doi: 10.1021/bi900907a

8. Faller P, Hureau C, Berthoumieu O (2013) Role of Metal Ions in the Self-assembly of the Alzheimer's Amyloid- β Peptide. *Inorg Chem* 52:12193–12206. doi: 10.1021/ic4003059
9. Zatta P, Lucchini R, Van R, Taylor A (2003) The role of metals in neurodegenerative processes: Aluminum, manganese, and zinc. *Brain Res Bull* 62:15–28. doi: 10.1016/S0361-9230(03)00182-5
10. Varadarajan S, Kanski J, Aksenova M, et al (2001) Different mechanisms of oxidative stress and neurotoxicity for Alzheimer's A β (1--42) and A β (25--35). *J Am Chem Soc* 123:5625–5631.
11. Kepp KP (2012) Bioinorganic Chemistry of Alzheimer's Disease. *Chem Rev* 112:5193–5239. doi: 10.1021/cr300009x
12. Butterfield DA, Lauderback CM (2002) Lipid peroxidation and protein oxidation in Alzheimer's disease brain: Potential causes and consequences involving amyloid β -peptide-associated free radical oxidative stress. *Free Radic Biol Med* 32:1050–1060. doi: 10.1016/S0891-5849(02)00794-3
13. Halliwell B (2007) Biochemistry of oxidative stress. *Biochem Soc Trans* 35:1147–1150. doi: 10.1042/BST0351147
14. Wang J, Xiong S, Xie C, et al (2005) Increased oxidative damage in nuclear and mitochondrial DNA in Alzheimer's disease. *J Neurochem* 93:953–962. doi: 10.1111/j.1471-4159.2005.03053.x
15. Hamley IW (2012) The Amyloid Beta Peptide: A Chemist's Perspective. Role in Alzheimer's and Fibrillization. *Chem Rev* 112:5147–5192. doi: 10.1021/cr3000994
16. Karran E, De S (2016) The amyloid cascade hypothesis: are we poised for success or failure? *J Neurochem* 139:237–252. doi: 10.1111/jnc.13632
17. Suberbielle E, Sanchez PE, Kravitz AV, et al (2013) Physiologic brain activity causes DNA double-strand breaks in neurons, with exacerbation by amyloid- β . *Nat Neurosci* 16:613–621. doi: 10.1038/nn.3356
18. Mayes J, Tinker-Mill C, Kolosov O, et al (2014) β -Amyloid Fibrils in Alzheimer Disease Are Not Inert When Bound to Copper Ions but Can Degrade Hydrogen Peroxide and Generate Reactive Oxygen Species. *J Biol Chem* 289:12052–12062. doi: 10.1074/jbc.M113.525212
19. Huang X, Atwood CS, Hartshorn MA, et al (1999) The A β Peptide of Alzheimer's Disease Directly Produces Hydrogen Peroxide through Metal Ion Reduction. *Biochemistry (Mosc)* 38:7609–7616. doi: 10.1021/bi990438f

20. Shearer J, Szalai VA (2008) The Amyloid- β Peptide of Alzheimer's Disease Binds CuI in a Linear Bis-His Coordination Environment: Insight into a Possible Neuroprotective Mechanism for the Amyloid- β Peptide. *J Am Chem Soc* 130:17826–17835. doi: 10.1021/ja805940m
21. Eskici G, Axelsen PH (2012) Copper and Oxidative Stress in the Pathogenesis of Alzheimer's Disease. *Biochemistry (Mosc)* 51:6289–6311. doi: 10.1021/bi3006169
22. Cho S-J, Roman G, Yeboah F, Konishi Y (2007) The road to advanced glycation end products: a mechanistic perspective. *Curr Med Chem* 14:1653–1671.
23. Münch G, Thome J, Foley P, et al (1997) Advanced glycation endproducts in ageing and Alzheimer's disease. *Brain Res Brain Res Rev* 23:134–143.
24. Janson J, Laedtke T, Parisi JE, et al (2004) Increased risk of type 2 diabetes in Alzheimer disease. *Diabetes* 53:474–481.
25. Imlay JA (2003) Pathways of Oxidative Damage. *Annu Rev Microbiol* 57:395–418. doi: 10.1146/annurev.micro.57.030502.090938
26. Genaro-Mattos TC, Dalvi LT, Oliveira RG, et al (2009) Reevaluation of the 2-deoxyribose assay for determination of free radical formation. *Biochim Biophys Acta BBA - Gen Subj* 1790:1636–1642. doi: 10.1016/j.bbagen.2009.09.003
27. Li KS, Rempel DL, Gross ML (2016) Conformational-Sensitive Fast Photochemical Oxidation of Proteins and Mass Spectrometry Characterize Amyloid Beta 1–42 Aggregation. *J Am Chem Soc* 138:12090–12098. doi: 10.1021/jacs.6b07543
28. Crescenzi O, Tomaselli S, Guerrini R, et al (2002) Solution structure of the Alzheimer amyloid β -peptide (1–42) in an apolar microenvironment. *Eur J Biochem* 269:5642–5648. doi: 10.1046/j.1432-1033.2002.03271.x
29. Yim HS, Kang SO, Hah YC, et al (1995) Free radicals generated during the glycation reaction of amino acids by methylglyoxal. A model study of protein-cross-linked free radicals. *J Biol Chem* 270:28228–28233.
30. Lederer MO, Klaiber RG (1999) Cross-linking of proteins by Maillard processes: characterization and detection of lysine-arginine cross-links derived from glyoxal and methylglyoxal. *Bioorg Med Chem* 7:2499–2507.
31. Wysocki VH, Resing KA, Zhang Q, Cheng G (2005) Mass spectrometry of peptides and proteins. *Methods* 35:211–222. doi: 10.1016/j.ymeth.2004.08.013
32. Chumsae C, Gifford K, Lian W, et al (2013) Arginine Modifications by Methylglyoxal: Discovery in a Recombinant Monoclonal Antibody and Contribution to Acidic Species. *Anal Chem* 85:11401–11409. doi: 10.1021/ac402384y

33. Ahmed MU, Brinkmann Frye E, Degenhardt TP, et al (1997) N-epsilon-(carboxyethyl)lysine, a product of the chemical modification of proteins by methylglyoxal, increases with age in human lens proteins. *Biochem J* 324:565–570.
34. Gersten RA, Gretebeck LM, Hildick-Smith G, Sandwick RK (2010) Maillard reaction of ribose 5-phosphate generates superoxide and glycation products for bovine heart cytochrome c reduction. *Carbohydr Res* 345:2499–2506. doi: 10.1016/j.carres.2010.09.013
35. Margoliash E, Frohwirt N (1959) Spectrum of horse-heart cytochrome c. *Biochem J* 71:570–572.
36. Choi S, Cooley RB, Voutchkova A, et al (2005) Oxidation of Guanosine Derivatives by a Platinum(IV) Complex: Internal Electron Transfer through Cyclization. *J Am Chem Soc* 127:1773–1781. doi: 10.1021/ja045194n
37. Lo TW, Westwood ME, McLellan AC, et al (1994) Binding and modification of proteins by methylglyoxal under physiological conditions. A kinetic and mechanistic study with N alpha-acetylarginine, N alpha-acetylcysteine, and N alpha-acetyllysine, and bovine serum albumin. *J Biol Chem* 269:32299–32305.
38. Fitch CA, Platzner G, Okon M, et al (2015) Arginine: Its pKa value revisited. *Protein Sci* 24:752–761. doi: 10.1002/pro.2647
39. Isom DG, Castañeda CA, Cannon BR, García-Moreno BE (2011) Large shifts in pKa values of lysine residues buried inside a protein. *Proc Natl Acad Sci U S A* 108:5260–5265. doi: 10.1073/pnas.1010750108
40. Triguero L, Singh R, Prabhakar R (2008) Comparative molecular dynamics studies of wild-type and oxidized forms of full-length alzheimer amyloid β -peptides A β (1–40) and A β (1–42). *J Phys Chem B* 112:7123–7131. doi: 10.1021/jp801168v
41. Zhao LN, Mu Y, Chew LY (2013) Heme prevents amyloid beta peptide aggregation through hydrophobic interaction based on molecular dynamics simulation. *Phys Chem Chem Phys* 15:14098–14106. doi: 10.1039/c3cp52354c
42. Walton DJ, Shilton BH (1991) Site specificity of protein glycation. *Amino Acids* 1:199–203. doi: 10.1007/BF00806917
43. Tomaselli S, Esposito V, Vangone P, et al (2006) The alpha-to-beta Conformational Transition of Alzheimer's A β (1–42) Peptide in Aqueous Media is Reversible: A Step by Step Conformational Analysis Suggests the Location of beta Conformation Seeding. *Chembiochem* 7:257–267. doi: 10.2210/pdb1z0q/pdb
44. Derrick JS, Lee J, Lee SJC, et al (2017) Mechanistic Insights into Tunable Metal-Mediated Hydrolysis of Amyloid- β Peptides. *J Am Chem Soc* 139:2234–2244. doi: 10.1021/jacs.6b09681

